Used Car Price Prediction

Phase I: Exploration and Linear Regression

October 20, 2026

1 Introduction

The goal of this project is to study the factors associated with used-car prices and compare several regression approaches. The response variable is car price in USD. The analysis moves from single-predictor regression to multiple regression, interaction effects, and a polynomial model, then compares the models using test-set error and regression diagnostics.

The working dataset is based on the Kaggle **Car Price Prediction Dataset for Regression Models**:

https://www.kaggle.com/datasets/dhruvnr/car-price-prediction-dataset-for-regression-models

The original working file contains 10,000 observations and 15 variables. Fuel Type, Location, and Insurance Type are excluded from the analysis.

2 Data Preparation

The main preprocessing steps are summarized below.

Item	Value
Original observations	10000
Original variables	15
Variables after removal	12
Missing values after removal	0
Duplicated rows	0
Observations used in analysis	5000

The retained categorical variables were converted to factors. Price was converted from INR to USD using the same conversion used in the working analysis.

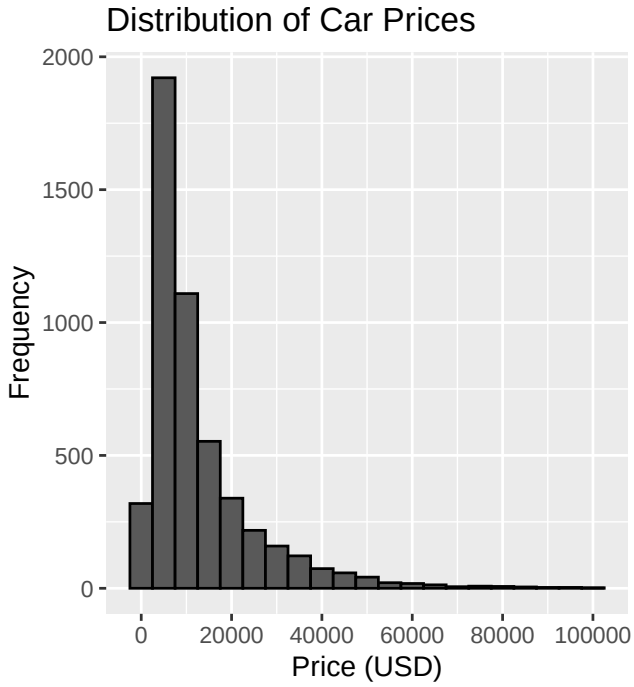
2.1 Quantitative Summary

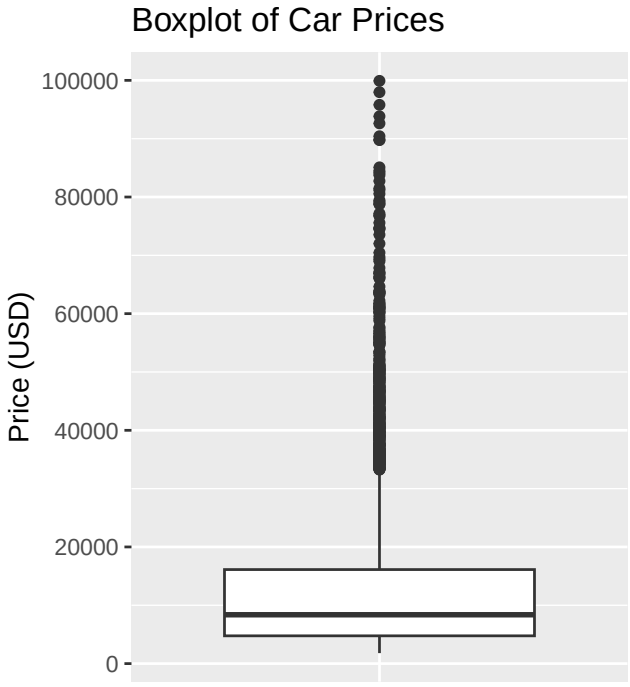
Table 1: Summary statistics for quantitative variables

	Min.Min.	Q1.1st Qu.	Median.Median	Mean.Mean	Q3.3rd Qu.	Max.Max.
Year	2000	2006.00	2012.0	2011.99	2018.00	2024
Odometer.Reading..km.	5043	52841.25	100890.5	101558.07	151698.75	199984
Number.of.Owners	1	1.00	2.0	2.00	3.00	3
Engine.Capacity..L.	1	2.00	3.0	3.03	4.00	5
Price.USD	1799	4760.25	8376.0	12715.51	16125.25	99915

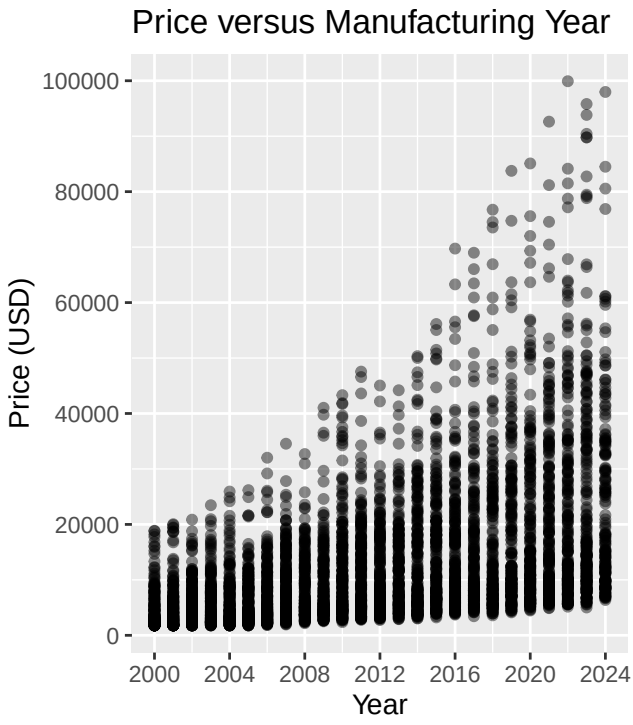
Car prices are strongly right-skewed: the mean price is higher than the median and a relatively small number of cars appear in the upper price tail.

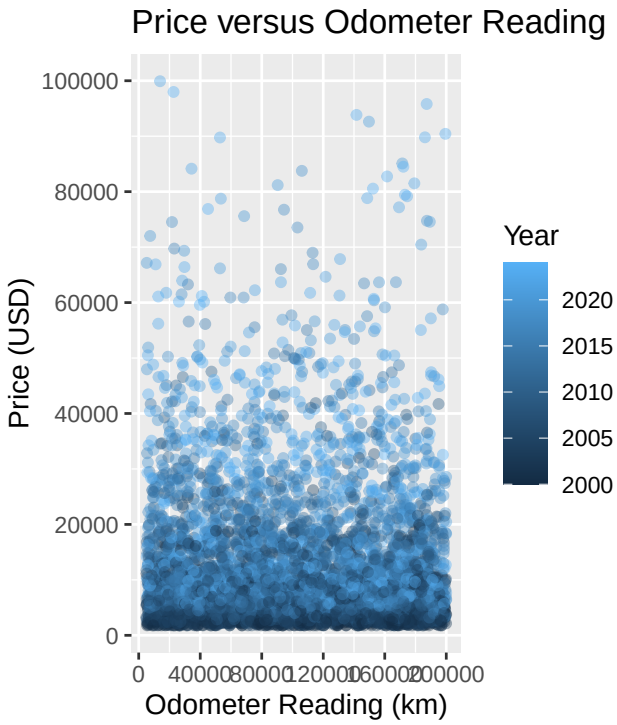
3 Exploratory Data Analysis



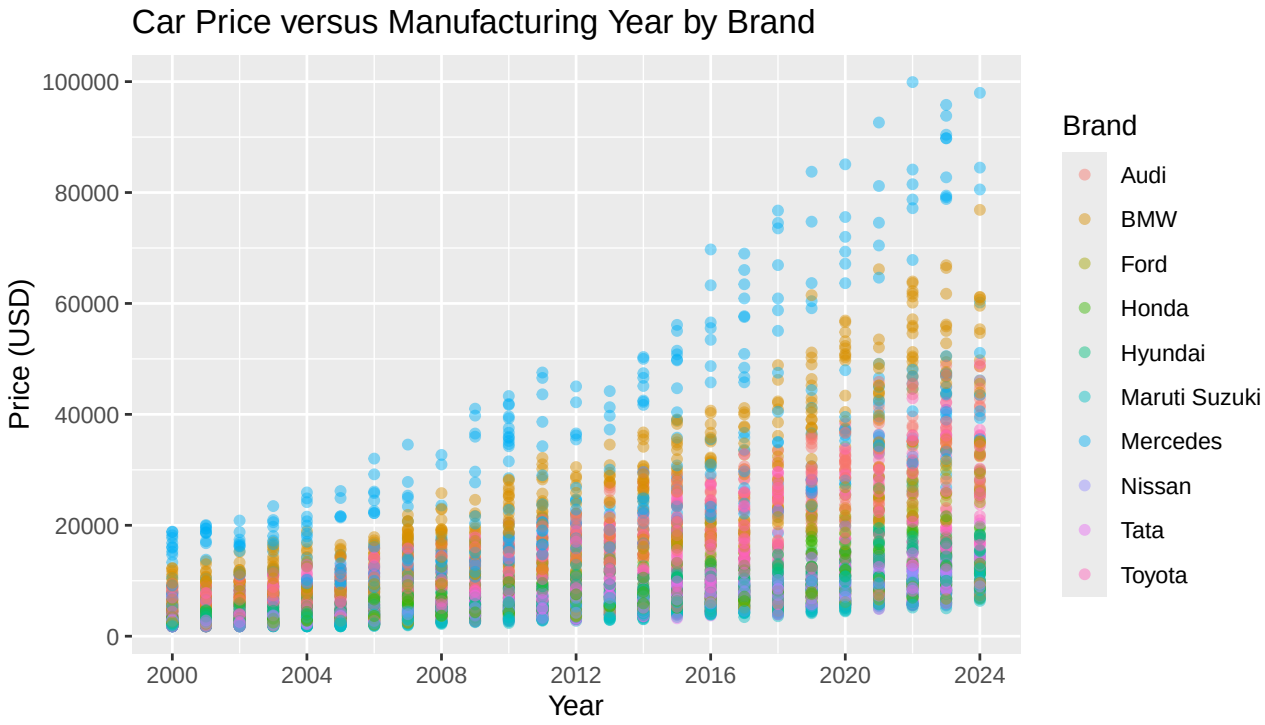


The histogram and boxplot confirm the long upper tail and the presence of many high-price statistical outliers.

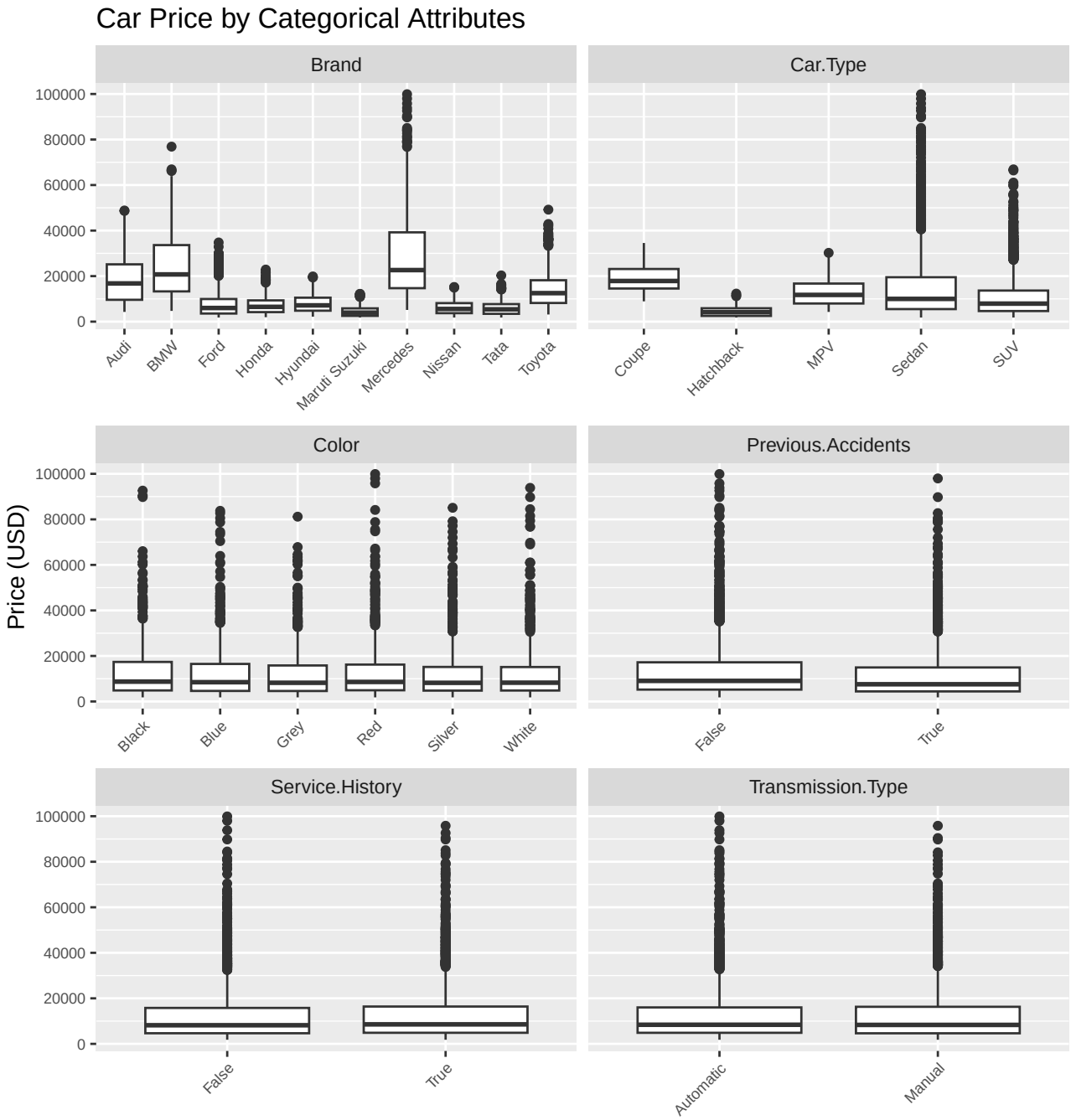




Manufacturing year shows a clear positive relationship with price. Odometer reading alone shows a much weaker pattern, and the color gradient indicates that year is an important part of the price structure.



The brand plot shows substantial differences in price level and suggests that the effect of year may vary across brands.



Brand and Car Type show the clearest price differences among the categorical variables. The other categorical variables overlap more strongly.

4 Hypotheses

The exploratory analysis motivates the following hypotheses:

1. Newer cars tend to have higher prices.
2. Higher odometer readings may reduce price, although the individual effect may be weak.
3. Prices differ across brands and car types.
4. The effect of year may depend on other car characteristics, motivating interaction terms.
5. The Year-price relationship may contain some nonlinearity, motivating a polynomial term.

5 Train/Test Split

Dataset	Observations	Percentage
Training	4000	80%
Test	1000	20%

All models are fitted using the training data and evaluated on the same test set.

6 Simple Linear Regression

Three separate simple linear regression models are first tested using Engine Capacity, Year, and Odometer Reading individually.

Table 2: Single-predictor regression comparison

	Predictor	Slope	p.value	R.squared	RMSE
Engine.Capacity..L.	Engine Capacity	-237.5634	0.1560	0.0005	13157.19
Year	Year	806.9047	0.0000	0.2207	11521.39
Odometer.Reading..km.	Odometer Reading	-0.0047	0.1772	0.0005	13139.51

Year is the strongest single predictor, explaining about 22% of the variation in price. Engine Capacity and Odometer Reading have almost no explanatory power when used alone. Even the Year-only model has a relatively high test RMSE, showing that a single predictor is not enough.

7 Multiple Linear Regression

A first MLR uses Year, Odometer Reading, and Brand.

Model	R.squared	Adjusted.R.squared	RSE	Test.RMSE
Manual MLR	0.657	0.656	7216.403	7633.752

Holding the other variables constant, Year has a strong positive coefficient, while Odometer Reading has a small negative coefficient. Brand also produces substantial price differences.

A broader MLR is then selected using stepwise AIC.

The selected main-effects model contains Brand, Year, Car Type, Previous Accidents, Number of Owners, Service History, Odometer Reading, and Transmission Type.

Table 3: Selected MLR: key coefficient estimates

	Predictor	Estimate	p.value
Year	Year	817.5553	0.0000
Previous.AccidentsTrue	Previous.AccidentsTrue	-2621.8663	0.0000
Number.of.Owners	Number.of.Owners	-828.8175	0.0000
Service.HistoryTrue	Service.HistoryTrue	901.4758	0.0000
Odometer.Reading..km.	Odometer.Reading..km.	-0.0061	0.0016
Transmission.TypeManual	Transmission.TypeManual	-354.8918	0.1052

The selected MLR substantially improves prediction compared with the single-predictor regressions.

8 Interaction Model

Two-way interactions are considered among the predictors selected in the MLR. The Brand x Car Type interaction is excluded because the initial all-pairwise specification created exact linear dependencies.

Table 4: Interactions retained by stepwise selection

Selected_Interaction
Brand:Year
Brand:Previous.Accidents
Year:Car.Type
Year:Previous.Accidents
Brand:Number.of.Owners
Brand:Transmission.Type
Year:Odometer.Reading..km.
Year:Number.of.Owners
Brand:Odometer.Reading..km.
Year:Transmission.Type
Year:Service.History
Number.of.Owners:Transmission.Type
Previous.Accidents:Service.History
Previous.Accidents:Transmission.Type
Car.Type:Number.of.Owners

The interaction model produces a large improvement in test error, indicating that the effect of one vehicle characteristic often depends on another.

9 Polynomial Regression

A quadratic Year term is added to the selected interaction model.

The quadratic term gives a smaller additional improvement beyond the interaction model.

10 Model Comparison

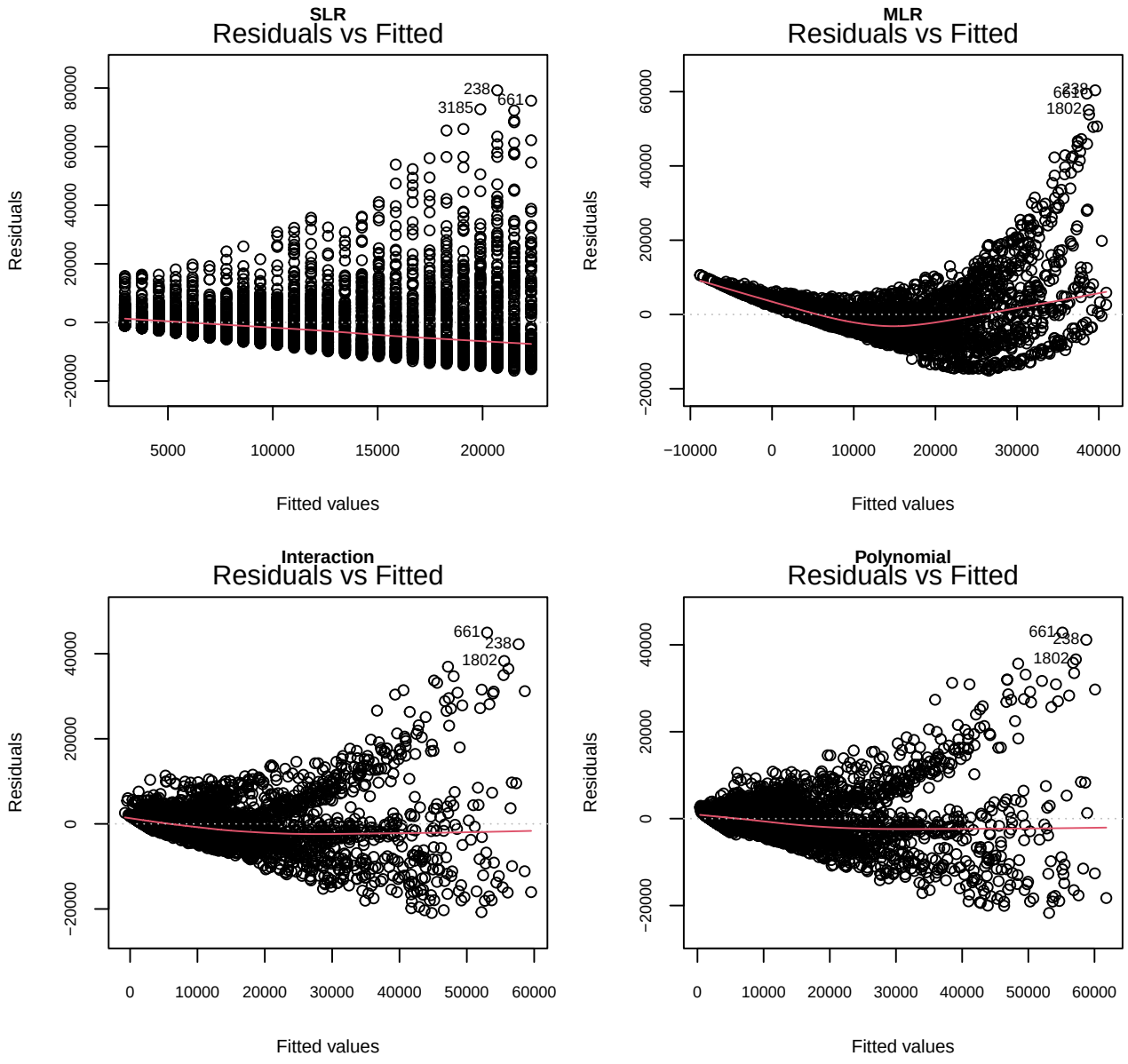
Table 5: Model performance comparison

	Model	R.squared	Adjusted.R.squared	RSE	MSE	RMSE
SLR	SLR	0.221	0.221	10865.529	132742481	11521.392
MLR	MLR	0.686	0.684	6913.205	55315456	7437.436
Interaction	Interaction	0.816	0.812	5332.672	33038637	5747.925
Polynomial	Polynomial	0.824	0.820	5222.444	31713776	5631.499

The polynomial model has the lowest test RMSE, while the largest improvement occurs when moving from the selected MLR to the interaction model. Predictive performance therefore improves as important interaction and nonlinear effects are introduced.

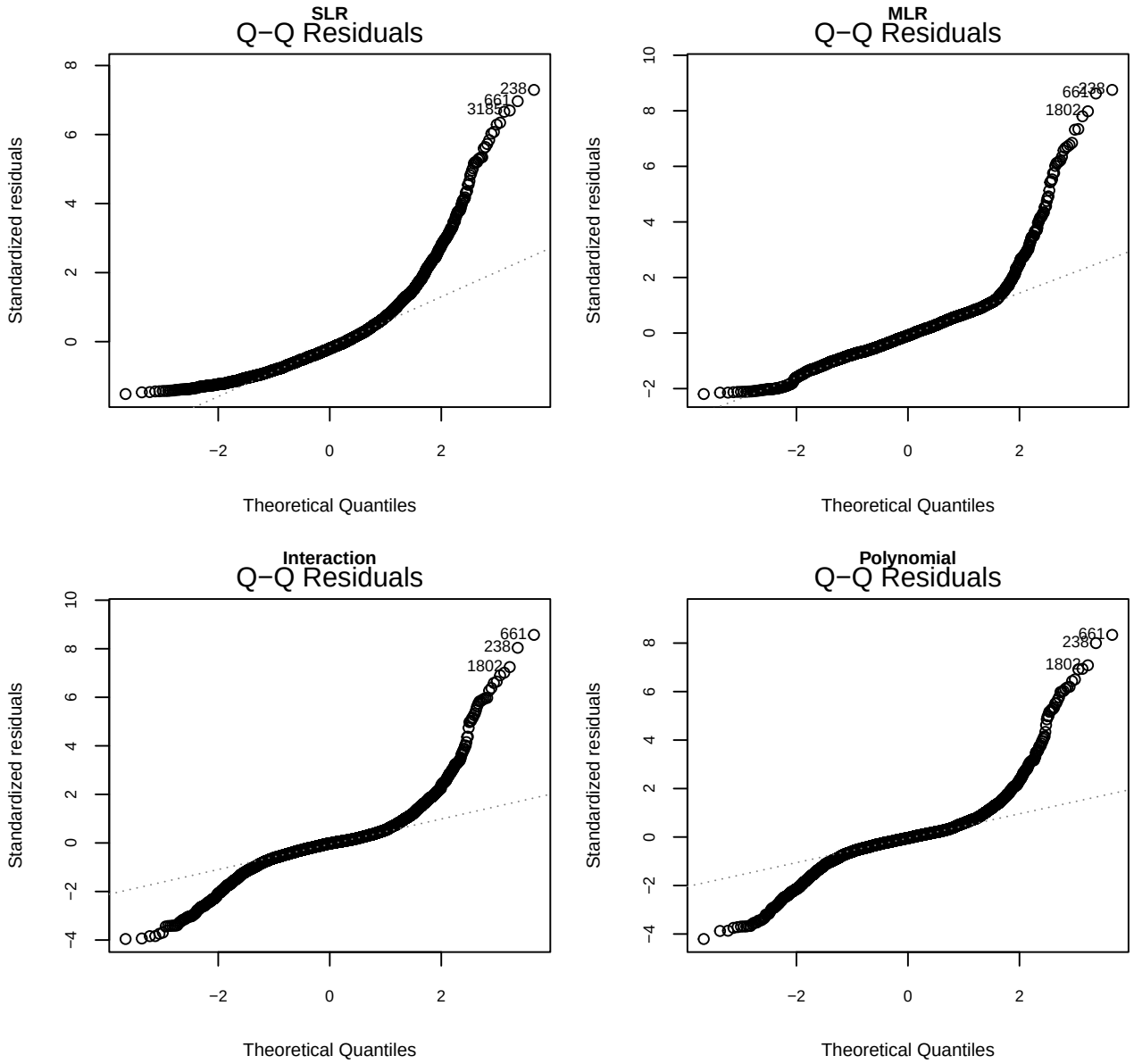
11 Regression Diagnostics

11.1 Residuals versus Fitted



The SLR and MLR retain substantial structure in the residuals. The interaction and polynomial models reduce this pattern, although residual spread still increases for higher fitted prices.

11.2 Normal Q-Q Plots



All models show departures from normality in the tails. The interaction and polynomial models fit the center of the residual distribution better, but heavy tails remain.

11.3 Outliers, Leverage, and Influence

Table 6: Diagnostic screening counts

	Model	Outliers	High_Leverage	Influential_Cooks
SLR	SLR	74	0	171
MLR	MLR	64	83	118
Interaction	Interaction	84	78	203
Polynomial	Polynomial	88	78	197

Table 7: Five most extreme standardized residuals in the polynomial model

	Observation	Standardized_Residual
661	661	8.339
238	238	8.001
1802	1802	7.085
3185	3185	6.938
207	207	6.916

In the final polynomial model, observations with large standardized residuals are present, but they are not automatically removed. Cook's distance and leverage are used to assess whether they dominate the fitted model.

The Cook's distance screening cutoff is 0.001, and 197 observations exceed it. The largest Cook's distance is 0.029, which is far below 1. The leverage cutoff is 0.041, and 78 observations exceed it.

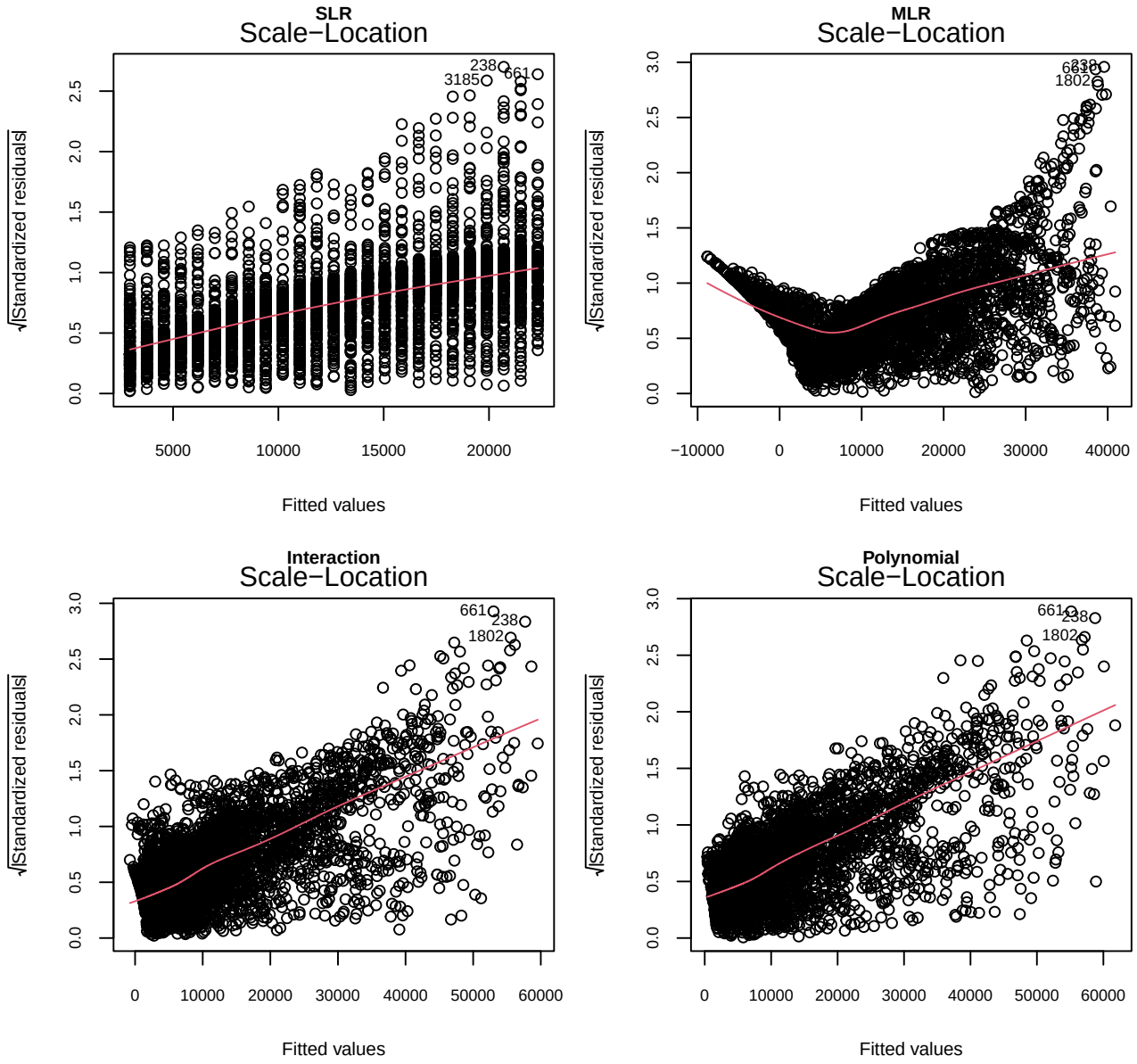
11.4 Multicollinearity

Table 8: VIF / GVIF results for the selected MLR

Predictor	GVIF	Df	GVIF^(1/(2*Df))
Brand	3.654	9	1.075
Year	1.015	1	1.008
Car.Type	3.671	4	1.177
Previous.Accidents	1.006	1	1.003
Number.of.Owners	1.003	1	1.001
Service.History	1.006	1	1.003
Odometer.Reading..km.	1.007	1	1.004
Transmission.Type	1.004	1	1.002

The adjusted GVIF values are close to 1, so the selected MLR does not show meaningful multicollinearity.

11.5 Scale-Location Plots



The Scale-Location plots confirm heteroscedasticity. Residual variability increases as fitted price increases, especially for the interaction and polynomial models.

11.6 Aliased Coefficients

Model	Aliased_Coefficients
Interaction	0
Polynomial	0

Both final models have zero aliased coefficients after excluding the problematic Brand x Car Type interaction.

12 Conclusion

The single-predictor regressions show that Year is the strongest individual predictor, while Engine Capacity and Odometer Reading alone explain very little of the variation in price. The selected MLR improves prediction substantially, and interaction terms provide the largest additional improvement. The polynomial model achieves the lowest test RMSE.

The diagnostics show that the final models are not perfect. Heteroscedasticity and heavy-tailed residuals remain, and some unusual observations are present. However, no single observation dominates the model according to Cook's distance, and the corrected interaction and polynomial specifications contain no aliased coefficients. Overall, the polynomial model gives the strongest predictive performance among the models evaluated in this Phase I analysis.